

MILAN SINGH

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PROFESSIONAL SUMMARY

Strategic AI Engineering Lead and Senior Data Scientist with 5 years of experience architecting autonomous data ecosystems and production-ready Generative AI solutions. Expert in scaling multi-agent frameworks (LangGraph/LangChain), high-precision Computer Vision systems, and automated Data Engineering pipelines. Proven track record in orchestrating digital transformations for high-growth startups, consistently delivering high-impact uplifts in ROI, user engagement, and operational efficiency through an AI-first strategic vision.

PROFESSIONAL EXPERIENCE

Senior AI Engineering Lead

Salty Accessories

Dec 2025 – Present

Delhi, India

- Architected a unified, autonomous data infrastructure on AWS (EC2, Data Lake), migrating from expensive third-party ETL tools to a custom Airflow-orchestrated ecosystem; deployed 35+ high-frequency pipelines (Shopify, Amazon, Myntra) within the first 60 days.
- Developed a Multimodal AI Image Recognition system for automated cataloging, utilizing Computer Vision to extract product attributes with 98% accuracy, reducing manual processing time from 20 minutes to <1 minute per SKU.
- Directing a cross-functional team of 4 software and data analysts while spearheading the strategic shift toward a data-driven, AI-first company culture.
- Developing an 'Agentic Commerce Protocol' leveraging LangChain-powered autonomous agents to enable customers to browse and purchase products via LLM-driven interfaces on both the website and mobile app.
- Engineered "Heartbeat," a gamified community engine for the mobile app, leveraging social competition mechanics to significantly uplift app downloads and viral engagement during peak sales periods.
- Built a unified financial intelligence suite providing founders and finance teams with a holistic view of P&L, automated expense/income tracking, and multi-channel ROI to optimize strategic spend and phase out underperforming platforms.
- Streamlined internal operations by automating manual tasks, including a regulatory-compliant barcode generation system and an intelligent label generator for sales channels like Blinkit, Myntra, and Shopify.

Data Scientist and AI Engineer

InHousen

Jan 2025 – Dec 2025

Netherlands (Remote)

- Engineered an end-to-end sales forecasting pipeline using XGBoost and LSTM, achieving ~90% accuracy by implementing a custom feature engineering process that captured complex seasonality and promotional impacts.
- Developed a proactive inventory management system that forecasted stockouts and recommended order quantities, reducing lost sales revenue and optimizing cash flow by preventing overstocking.
- Architected an AI agentic framework using LangChain and LangGraph for hyper-personalized outreach, boosting reply rates from 2% to 10% and significantly shortening the client's sales cycle.
- Implemented a RAG-based AI chatbot providing stakeholders with on-demand, natural language access to critical business intelligence, reducing reliance on the data team for ad-hoc reports.
- Designed an AI-driven lead scoring model analyzing LinkedIn data to rank candidates by their probability of engagement, enhancing the relevance and success of recruitment outreach.

Data Science Engineer

Pintel.ai

July 2024 – Dec 2024

Bengaluru, India

- Led development of AI-powered personalized message sequences, boosting open rates by ~30% by leveraging NLP to extract and incorporate insights from social media activity into outreach copy.
- Built and deployed end-to-end AI solutions for B2B clients, automating the aggregation of firmographic and technographic data to identify high-intent target accounts.
- Created AI-driven tools for lead filtering, qualification, and scoring, improving lead management efficiency and helping clients target high-value prospects more effectively.
- Managed the full-cycle execution of personalized campaigns for over 6,000 leads, ensuring seamless data flow from filtering and scoring through to automated delivery.

Senior Data Analyst

Skidos Learning Labs

Aug 2021 – July 2024

Delhi, India

- Orchestrated a rigorous A/B testing framework, leading to a ~15% UX improvement by validating changes to the user onboarding flow and subscription funnels.
- Rolled out a weekly Learning Report that increased user engagement by ~25% and comprehension of weekly progress by 30%.
- Analyzed pre- and post-SDK releases by monitoring performance and stability metrics, driving a ~15% improvement in key business outcomes and enabling over 10 successful feature launches.
- Segmented high-value users using RFM analysis and clustering models, informing the development of targeted features that boosted engagement and retention by ~15%.
- Worked on ETL pipelines for 10+ projects, ensuring high reliability and uptime for business-critical data.
- Delivered over 10 in-depth analyses, uncovering insights that improved product quality, user experience, and overall business growth.

TECHNICAL EXPERTISE

Artificial Intelligence: Generative AI, LangChain, LangGraph, RAG, Computer Vision (OCR, CLIP), XGBoost, LSTM, NLP

Data Engineering: Apache Airflow, AWS Data Lake, ETL/ELT, SQL, dbt, Data Modeling, Pandas, NumPy

Cloud & Architecture: AWS (EC2, S3, Athena, Lambda), Google Cloud (GCP), MongoDB, FastAPI, Node.js

Strategy & Growth: A/B Testing, RFM Analysis, ROI Optimization, Power BI, Tableau, WebEngage, Jira

Languages: Python (Expert), SQL (Expert), JavaScript, HTML/CSS

EDUCATION

Guru Ghasidas University

Bachelor of Technology in Chemical Engineering

CG, India

Aug 2017 – June 2021